

Multiple Choice (10 marks)

1. What unusual feature of the cycads makes them a remarkable group among the seed plants?
 - (a) They have flowers that bloom for several years before producing seeds
 - (b) They have a unique leaf structure
 - (c) They can survive in extremely cold climates
 - (d) They produce large seeds
 - (e) Their seeds can remain dormant for up to a century before germinating
2. Describe the function of the lateral-line system in fishes.
 - (a) The lateral-line system in fishes helps with the digestion of food.
 - (b) The lateral-line system in fishes is used for excreting waste materials.
 - (c) The lateral-line system in fishes is involved in the secretion of hormones.
 - (d) The lateral-line system in fishes is used for breathing.
 - (e) The lateral-line system in fishes is used for vision.
3. Which of the following would maintain a constant body temperature in spite of changes in the environmental temperature: frog, robin, fish, dog, or lizard?
 - (a) dog, robin, fish
 - (b) robin, dog
 - (c) lizard, frog
 - (d) fish, robin
 - (e) fish, frog
4. Compare the methods of obtaining oxygen in the earthworm, the frog, the fish and man.
 - (a) Earthworms use gills, frogs use both gills and lungs, fish use lungs, and humans use skin.
 - (b) Earthworms and humans use lungs, frogs and fish use gills.
 - (c) All use gills for obtaining oxygen.
 - (d) Earthworms and humans use gills, frogs use skin, and fish use lungs.
 - (e) Earthworms use lungs, frogs use gills, fish use skin, and humans use gills.
5. Explain how an estimate of the age of a rock is made on the basis of the radioactive elements present.

- (a) By assessing the fossil content found within the rock layers
- (b) By comparing the density of the rock to a known age database
- (c) By examining the color and texture of the rock
- (d) By counting the layers of the rock
- (e) By measuring the amount of carbon-14 remaining in the rock

True / False (10 marks)

Answer True or False

- 6. True or False: Insulin decreases blood sugar levels by promoting the breakdown of glycogen in the liver.
- 7. True or False: Sexual dimorphism is primarily caused by artificial selection, which is driven by human preference for specific traits.
- 8. True or False: Adenosine triphosphate functions primarily as an energy currency and does not act as a second messenger for calcium release.
- 9. True or False: Complete equilibrium in a gene pool is not expected due to ongoing evolutionary processes and environmental changes.
- 10. True or False: An increase in solar radiation is a biotic factor that enhances the growth rate of all populations in an ecosystem.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss how natural selection may lead to the rapid emergence of drug-resistant HIV strains following treatment with 3 TC, including the role of pre-existing viral populations.
- 12. Discuss the evolutionary advancements found in Selaginella that differentiate it from ferns, focusing on vascular structures and reproductive strategies.

End of Paper